Sources of lifetime inequality revisited

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Motivation

- Initial conditions have great impact in determining lifetime earnings, consumption and welfare (Huggett, Ventura and Yaron, 2011).
- Important for insurance purposes.
- Initial conditions are somewhat determined by parental background.
- They are important not only for inequality, but also to determine social mobility.

Research Questions

- Many ways initial conditions matter. Focus on one particular mechanism: human capital accumulation.
- How much initial conditions matter? Which conditions have the greatest quantitative impact?
- How much does parental background/wealth influence in determining the initial conditions and lifetime welfare?
- What is the impact of the rise of college tuition and skill-biased technical change?

Motivation

- Financial resources matter to accumulate human capital.
- Borrowing constraint might bind and agent may not accumulate human capital optimally.
- Intergenerational transmission of wealth can alleviate the borrowing constraints for wealthier households.
- There is also well documented intergenerational transmission of abilities/ human capital, which also matters for determining lifetime welfare.

This paper

- Develop a quantitative model, with inter vivos wealth transfers and parental altruism, to study the evolution of inequality for a cohort, focusing on human capital and financial frictions.
- Quantitatively assess the importance of initial conditions: ability and initial wealth.
- Assess the importance of parental background and transfers, and their implication for intergenerational mobility.

Preliminary Results

- Initial conditions have a great impact in determining lifetime welfare. Almost 10% of variance in lifetime income and wealth is explained by initial conditions.
- Initial wealth inequality affects lifetime welfare and income. Accounting for intergenerational correlation of abilities is also important.
- Both wealth and ability have an important impact to determine lifetime welfare

Literature Review

- **Human Capital:** Becker (1964, 1979), Becker and Tomes (1979, 1986), Ben-Porath (1967).
- **Earnings Dynamics over the life cycle:** Huggett, Ventura and Yaron (2006, 2011), Guvenen (2009), Guvenen and Kuruscu (2010), Heathcote et. al. (2005), Meghir and Pistaferri (2011)
- **Intergenerational Transmission of wealth:** De Nardi (2004), Abbott et. al. (2018),
- **Costly Human capital accumulation and intergenerational mobility:** Loury (1981), Laitner (1992)
- **Human capital and borrowing constraints:** Lochner, Monge-Maranjo (2011), Hai and Heckman (2017)

Outline

The model

Empirical Analysis

Calibration

Results and Discussion

Conclusions

The model

Demographics, preferences and heterogeneity

- Partial equilibrium, OLG, life cycle and finite lives.
- Three stages of life cycle: education, work and retirement. During work stage, agent have a kid to whom they will transfer part of their wealth in the future.
- Parents are altruistic with their children. Agents maximize their present discounted value of utility flow and they weight their offspring lifetime utility.
- Maximize $\mathcal{U}_k = U_k + \gamma U_{k+1}$, where $U_k = \sum_{t=1}^{T} u(c_t^k)$ is the utility from consumption of generation k.
- Heterogeneity: Agents are heterogeneous in three aspects: (i) innate ability θ, (ii) financial wealth a and human capital h. Ability θ is fixed throughout lifetime, whereas agents can accumulate financial wealth and human capital.
- Government levies taxes and provides pension.

Human capital and earnings

- During working stage agents accumulate human capital with a Ben-Porath production function.
- Agents receive idiosyncratic human capital shocks every period.
- Human capital depreciates at rate δ per period.
- Human capital production function

$$h_{t+1} = \exp(z_{t+1})(h_t(1-\delta) + \theta(s_t h_t)^{\alpha})$$

- Earnings determined by $wh_t(1-s_t)$, where w depends on college decision.

The government and borrowing constraints

- The tax system has two parts: a proportional labor earnings tax to finance the social security system and a progressive tax schedule.
- The social security tax is a constant flat rate τ_{ss}. During retirement agents receive a constant pension p
- The progressive tax schedule is as in Benabou (2002). A worker with labor earnings y has a disposable income equals to $y^d = \mu y^{1-\tau}$, where τ represents the progressivity of the tax system.
- Two different borrowing restrictions: no borrowing and ad-hoc credit constraints.

Intergenerational links

- Agents have to choose how much resources to transfer to their offspring.
- Parents transfer wealth to their kids.
- Innate ability is correlated across generations and follows this process:

$$\log(\theta_{n+1}) = (1-\rho)\log(\bar{\theta}) + \rho\log(\theta_n) + \varepsilon$$

  where $\varepsilon \sim N(0, \sigma_\varepsilon^2)$
- Initial ability determines initial human capital.

Timeline

- In the model, life cycle starts at age 18 and agents decide whether to attend college depending on initial wealth.
- They work, have a kid, and save for self-insurance, smooth consumption, and accumulate wealth to make a transfer to the kid.
- Make an inter-vivos transfer when kid has to decide whether to go to college.
- Retire at age 65. Die at 80.

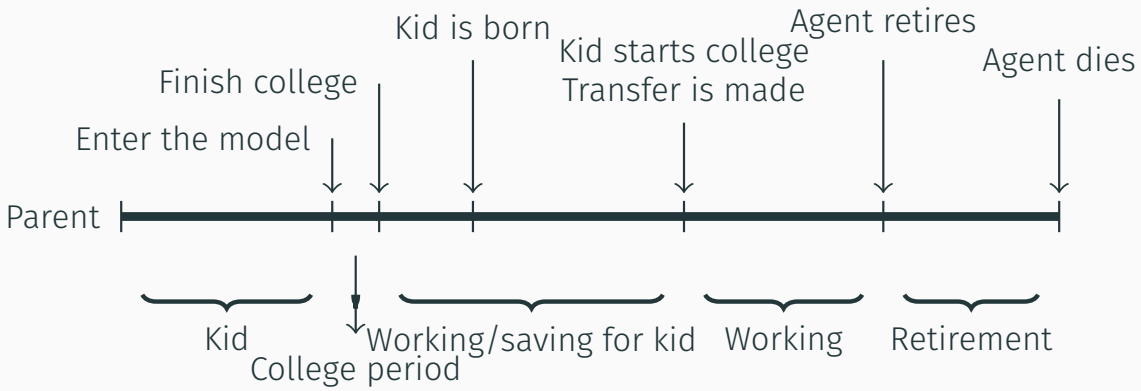


Figure 1: Timeline

Life cycle: Education

- Agents can invest in education (go to college) and work in the high skilled sector with a wage w_h or not to go to college and work in the low skilled sector an earn a wage w_l, for each unit of human capital they provide in the market
- Going to college costs P_c units of consumption good. The Bellman equation for a worker who decides whether to go to college is

$$V(a, h; \theta) = \max\{V^c(a, h; \theta); V^{nc}(a, h; \theta)\}$$

The Bellman equation of a college student is

$$V_j(a_j, h_j; \theta) = \max_{c_j, s_j} u(c_j) + \beta \mathbb{E}_z[V_{j+1}(a_{j+1}, h_{j+1}; \theta)]$$

$$\text{s.t.} \qquad c_j + a_{j+1} = a_j(1 + r) - P_c$$

$$h_{j+1} = (1 - \delta)h_j + \theta(h_j)^\alpha$$

Life cycle: Work

- Agents accumulate wealth for two reasons: 1) self-insurance and consumption smoothing and 2) transfer wealth to their offspring.
- Agents can use their time in accumulating human capital or provide hours in the market.
- The Bellman equation for a worker hired at wage w

$$V_j(a_j, h_j; \theta) = \max_{c_j, s_j} u(c_j) + \beta \mathbb{E}_z[V_{j+1}(a_{j+1}, h_{j+1}; \theta)]$$

$$\text{s.t.} \quad c_j + a_{j+1} = a_j + \lambda(ra_j + (1 - \tau_{ss})wh_j(1 - s_j))^{1-\tau}$$

$$h_{j+1} = \exp(z_{t+1})((1 - \delta)h_j + \theta(s_j h_j)^{\alpha})$$

Life cycle: Transfer

The Bellman equation of a parent who decides how much wealth to transfer to his offspring is

$$V(a_j, h_j; \theta) = \max_{c_j, s_j, tr} u(c) + \beta \mathbb{E}_z[V(a_{j+1}, h_{j+1}; \theta)] + \gamma \mathbb{E}_{\theta'} \left[V^o(tr; \theta')\right]$$

$$\text{s.t.} \quad c_j + a_{j+1} + tr = a_j + \lambda(ra_j + (1 - \tau_{ss})wh_j(1 - s_j))^{1-\tau}$$

$$h_{j+1} = \exp(z_{t+1})((1 - \delta)h_j + \theta(s_j h_j)^{\alpha})$$

After the transfer is made, the problem is the same as in the work periods.

Life cycle: Retirement

The Bellman equation of a retiree is

$$V(a_j) = \max_{c_j} + \beta V(a_{j+1})$$

$$\text{s.t.} \qquad c_j + a_{j+1} = p + (1+r)a_j$$

Equilibrium

The Steady State Competitive Equilibrium in this economy is a sequence of policy functions $\{c_j, s_j\}_{j=1}^{J}$, value functions $\{V_j\}_{j=1}^{J}$ and measures over states $\{g(a, h; \theta)\}_{j=1}^{J}$ such that, given prices r, w_l, w_h, the government policy and the initial distribution $g_0(a, h; \theta)$:

- Given prices and government policies, policy functions and value functions solve individual's problem.
- Measures $\{g(a, h; \theta)\}_{j=1}^{J}$ are consistent with policy functions and shocks.

Empirical Analysis

Empirical Analysis

- Data sources: PSID, NLSY79, NLSY79-Child/YA, NLSY97.
- Income profiles and inequality measure in the life cycle for a cohort.
- Income profiles by education level.
- Intergenerational correlation of abilities
- Intergenerational persistence of education.
- Education and wealth.
- Education and parental transfers.

Data

- Panel Study of Income Dynamics
  - First interviews in 1968. It samples more than 18,000 individuals from 5,000 families in the US. Representative of the US. Annual waves until 1997 and biannual since then.
  - Detailed data for household head and wife, some details about kids.
  - Once kids leave home and form their own household unit, they are kept in the survey as a separate household unit.
  - I use these data to compute statistics throughout lifetime.
- National Longitud Survey of Youth
  - Two cohorts, 1979 and 1997 (NLSY79, NLSY97). Approximately 12.000 and 9.000 individuals surveys in their first rounds, respectively.
  - Surveys youth between 14 and 22 years old in the first round. It follows those individuals since then.
  - I use these data to compute income profiles and compute intergenerational transfers from parents.

Income profiles and inequality measures over the life cycle

- Focus on male, household heads from PSID.
- Each age is a bin of 5 years with center in age.
- Statistical model

$$\text{stat}_{i,j} = \text{age effect}_i + \text{time effect}_j + \text{cohort effect}_c + \varepsilon_{i,j} \tag{1}$$

- Focus on time effect.

Income profile over the life cycle

Figure 2: mean earnings and Gini by age

Income profile by education level: PSID

- Analyze two separate sub samples. Non-college and college educated people.

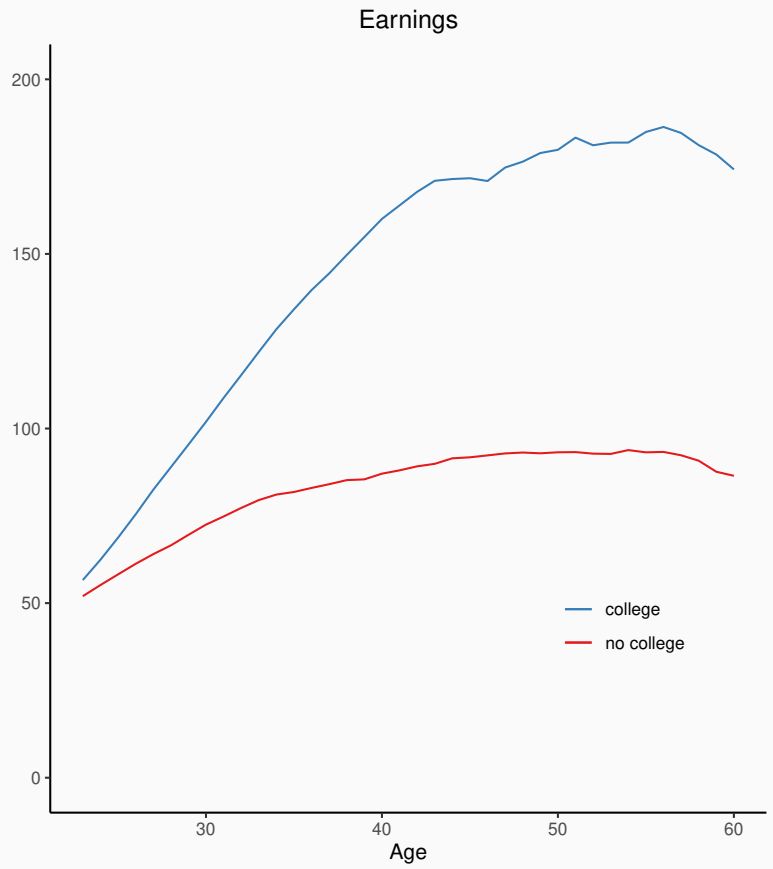


Figure 3: Income profile by education level: PSID

Intergenerational correlation of abilities

- Data from the NLSY79 and the NLSY79-Children and Young adults.
- AFQT percentile score as a measure of ability for parents and PIAT scores for children.

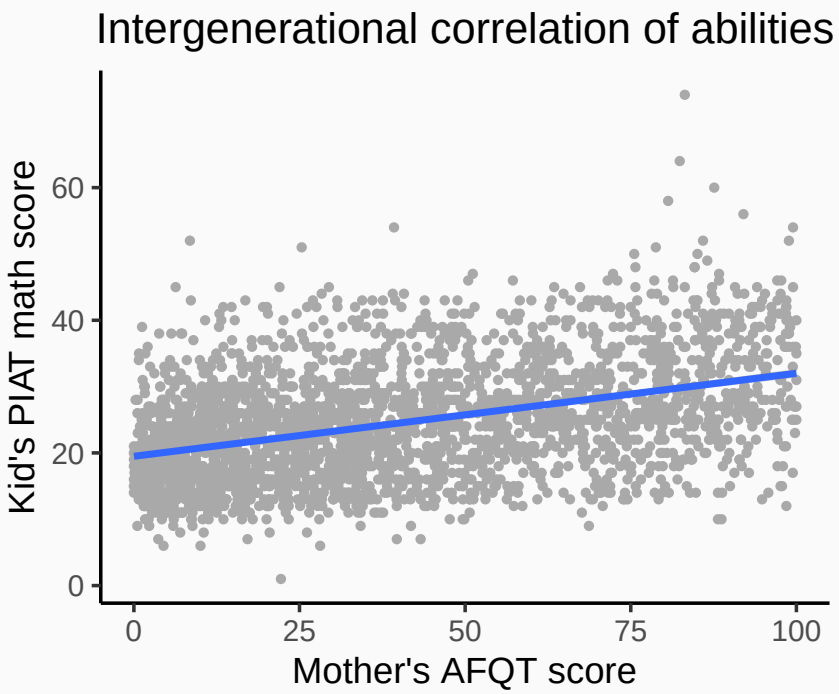


Figure 4: Intergenerational correlation of abilities, math scores

Intergenerational persistence of education

- Information about parent's education.
- Years 1974-2015.
- Focus on individuals aged 23-27.
- Probability of attending college, conditioning on parent's education.

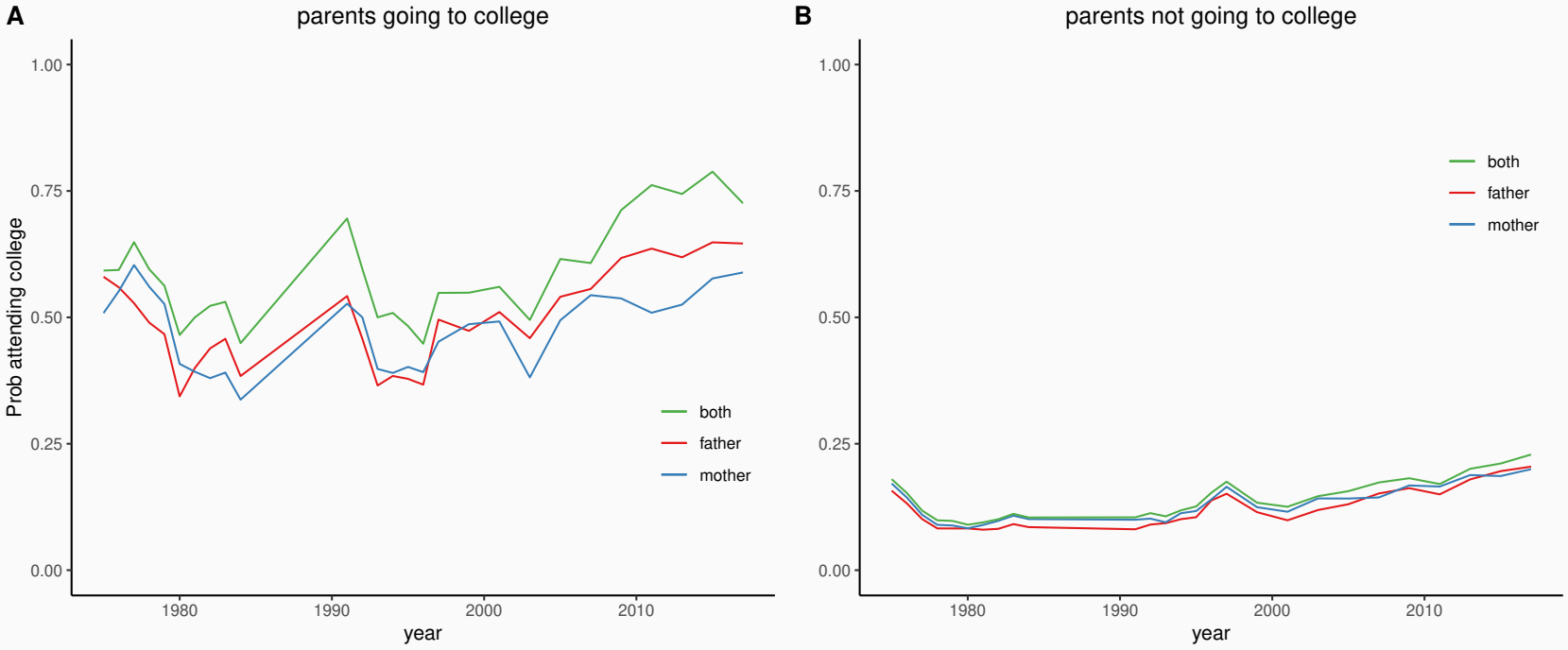


Figure 5: Probability of attending college, given parent's education

Education and wealth

- Wealth data from PSID. Years 1984, 1989, 1994 y biannual waves between 1999 y 2015.
- Construct three different wealth classes (bottom, middle, top), according to their relative positions with respect to the rest of individuals in the same cohort.
- Probability of attending college, conditional on wealth class.
- Results are robust to the inclusion of housing in the definition of wealth.

Education and wealth

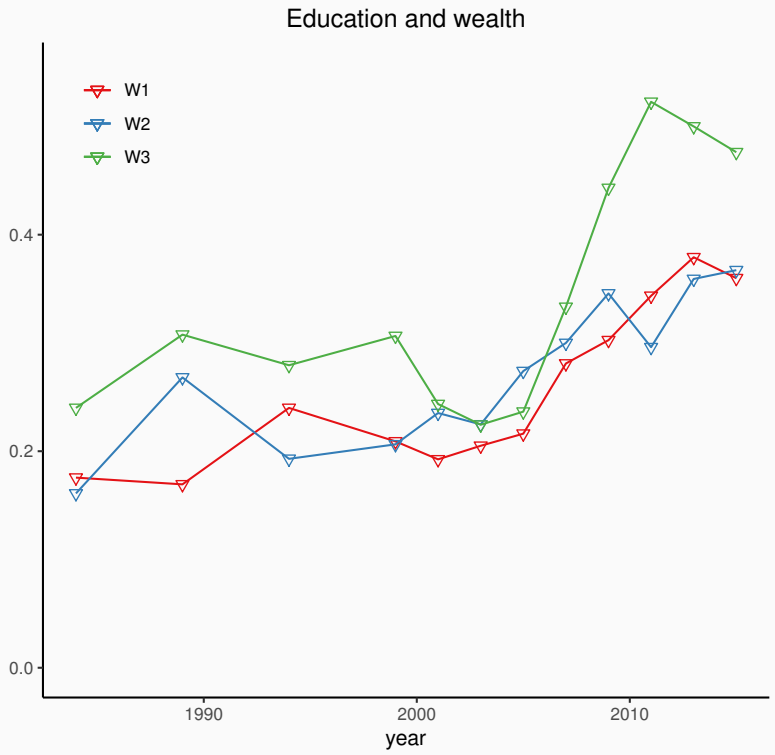


Figure 6: Probability of attending college conditional on wealth class

Education and parental transfers

- Data from NLSY97, cohorts 1980 and 1981.
- Descriptive Statistics.
- Construct “total transfers from parents” as all the transfers made from parents to kids during ages 16 and 22, without considering allowances. Estimates of transfers when the actual sum is not available.
- Results from three models: 1) Linear probability model, 2) Logit model.

Education and parental transfers: descriptive statistics

Cohort	Q1	Median	Mean	Q3	Fraction Tr >0
1980	300	750	3493	2200	39.4%
1981	300	850	3292	2138	39.9%

Table 1: Descriptive statistics transfers

Education and parental transfers: Histogram

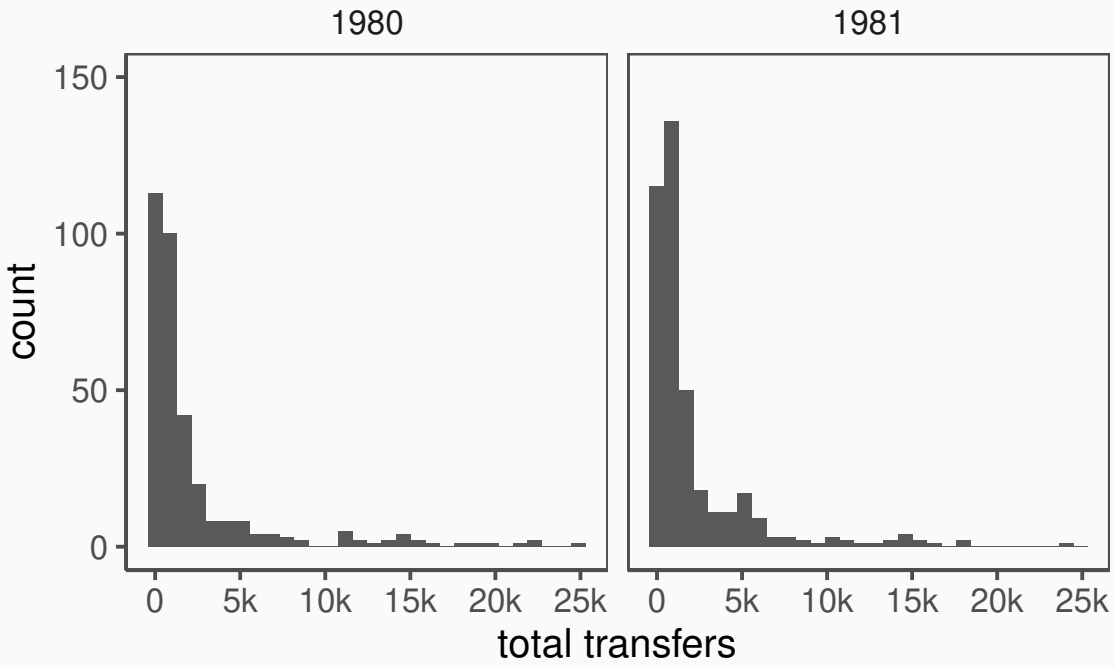


Figure 7: Histogram of positive total transfers

Education and parental transfers: Logit

	Dependent variable:			
	educ			
	(1)	(2)	(3)	(4)
received	1.019***	0.994***	0.757***	0.927***
	(0.104)	(0.105)	(0.121)	(0.140)
gender		0.461***	0.611***	0.601***
		(0.105)	(0.120)	(0.139)
race		0.238***	0.107*	0.091
		(0.051)	(0.060)	(0.076)
log(fam_income)			0.509***	0.363***
			(0.107)	(0.138)
fam_wealth			0.003***	0.003***
			(0.0005)	(0.001)
college_res_p				0.890***
				(0.175)
Constant	−1.288***	−2.791***	−4.834***	−4.462***
	(0.073)	(0.255)	(0.461)	(0.614)
Observations	1,867	1,867	1,635	1,139
Log Likelihood	−1,096.043	−1,075.214	−859.347	−626.856
Akaike Inf. Crit.	2,196.087	2,158.428	1,730.695	1,267.712

Calibration

Calibration

Parameter	Meaning	Value	Source/Target
		Prices	
r	rate of return	4%	calibrated
w_l	low wage	1	normalization
w_h	high wage	1.6	(Goldin and Katz (2007))
p	pension	40.	40% of mean earnings
P_c	price of college education	40.	share college educated agents (30%)
		Preferences and human capital	
β	Discount Factor	0.95	Standard
σ	CRRA utility function	2	Standard
α	production function	0.5	HVY (2011)
γ	Parameter of Altruism	0.64	(calibrated)
δ	Human capital depreciation	0.01	(calibrated)
σ_z	Variance human capital shocks	0.136	(calibrated)
		Government	
μ	Tax constant	0.902	Guner et. al (2014)
τ	Tax progressivity	0.036	Guner et. al (2014)
τ_{ss}	Social security tax	0.106	HVT(2011)

Table 3: Calibración

Model Fit: Income Profile

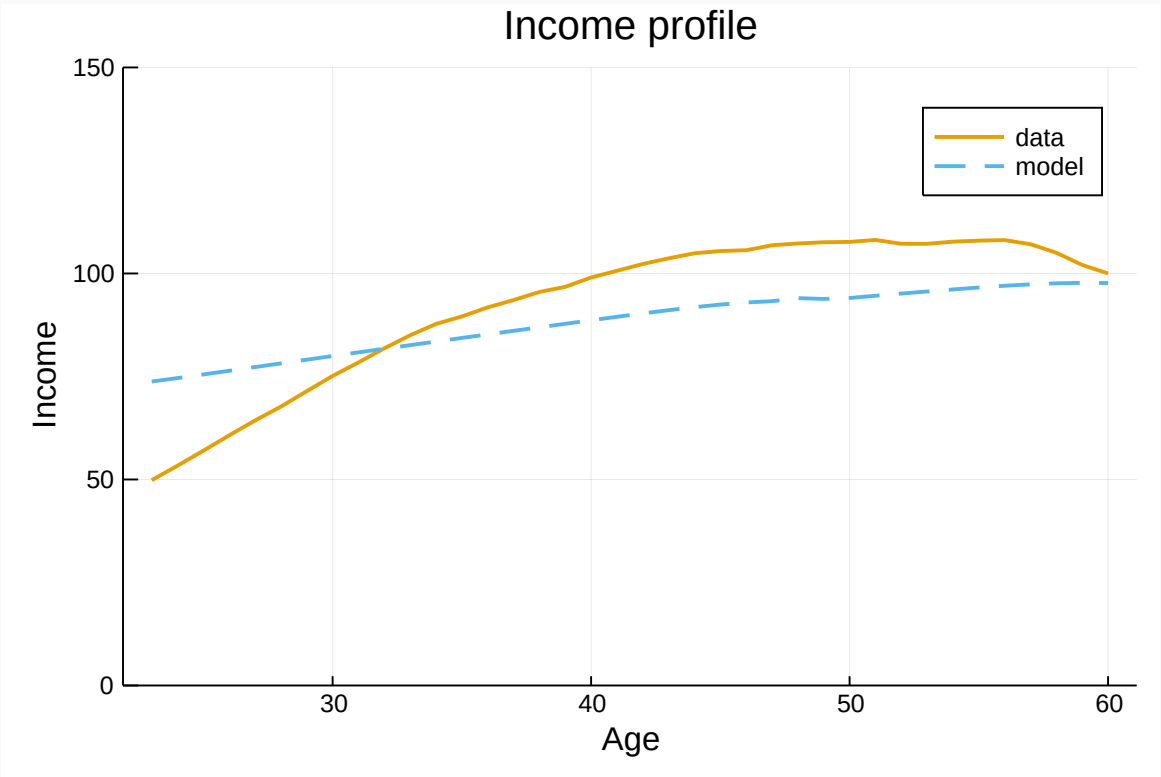
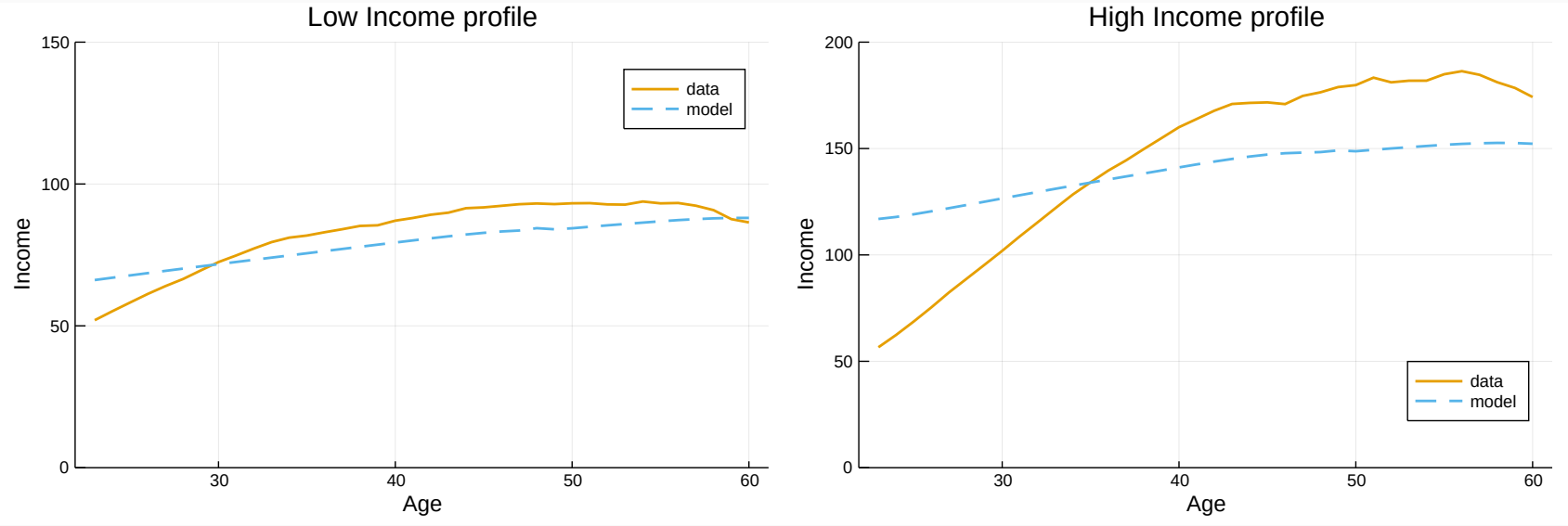


Figure 8: Model and Data: income profile

Model Fit: Income Profile by education



(a) low income profile

(b) high income profile

Figure 9: Model and Data: profiles

Model Fit: Gini

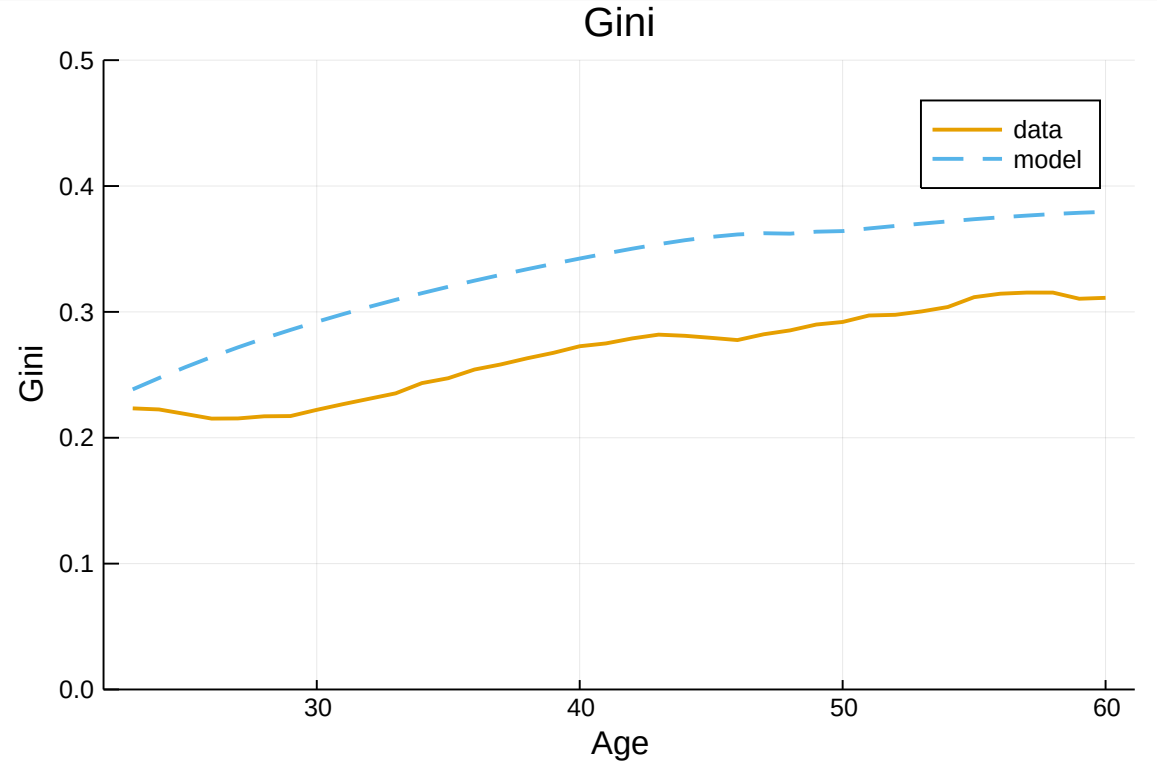


Figure 10: Model and Data: Gini index

Model Fit: Initial wealth/ transfer

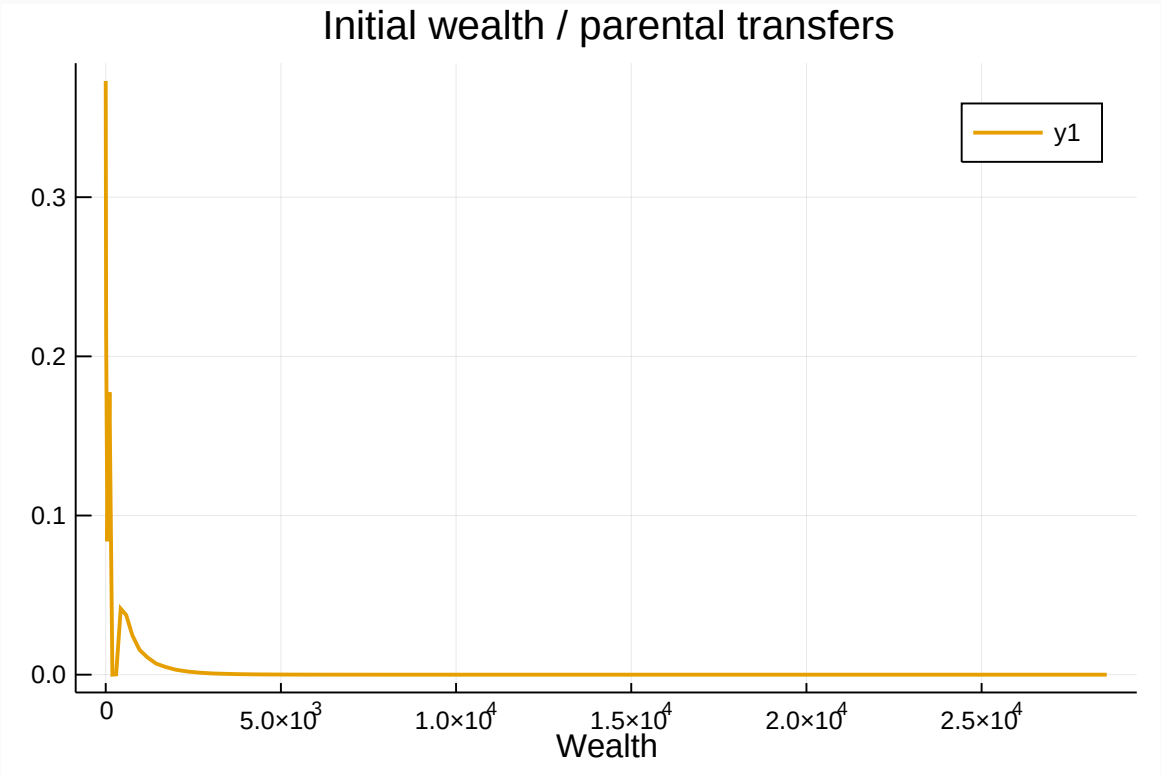


Figure 11: Initial wealth

Model fit: Other moments

Moment	Target	Model
P(college|no college)	0.2	0.05
P(college|college)	0.5	0.38
replacement rate	0.4	0.45
share of college students	0.3	0.15
share of agents with zero transfers	0.6	0.37
β intergenerational income elasticity	0.4	0.43

Table 4: Model Fit

Results and Discussion

Sources of Lifetime Inequality

Compare the present discounted value of income, wealth and utility by changing 1) the level initial wealth, 2) innate ability.

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, θ_l)	-7.37%	-14.57%	-14.5&
(a_m, θ_h)	10.48%	20.13 %	20.03%
(a_l, θ_m)	-1.32%	-1.03%	-1.51&
(a_h, θ_m)	8 %	1.39 %	6.01%

Table 5: Lifetime Inequality

Variance Decomposition in Dynamic Programming

The present discounted value of utility can be written as

$$B = \sum_{t=0}^{T} \beta^t u(x_t) \tag{2}$$

where x_t is the state at time t, which may follow a Markov process. The value function is define as the sum of expected discounted future pay offs, given the initial state x

$$J(x) = \mathbb{E}(B|x_0 = x) \tag{3}$$

It is useful to define the following operator

$$M(x) = \mathbb{E}(B^2|x_0 = x) \tag{4}$$

Variance Decomposition in Dynamic Programming (2)

Notice that $J(x)$ and $M(x)$ can be written as

$$J(x) = u(x) + \beta P(x'|x)J(x') \tag{5}$$

$$M(x) = u(x)^2 + 2\beta u(x)P(x'|x) + \beta^2 P(x'|x)M(x') \tag{6}$$

The previous two equations can be computed recursively. Using $J(x)$ and $M(x)$ we can compute the variance decomposition of a variable

$$\mathrm{Var}(Y|x) = \mathbb{E}(Y^2|x) - \mathbb{E}(Y|x)^2 = M(x) - J(x)^2 \tag{7}$$

$$\mathrm{var}(Y) = \underbrace{\mathbb{E}(\mathrm{var}(Y|X))}_{\text{due to shocks}} + \underbrace{\mathrm{var}(\mathbb{E}(Y|X))}_{\text{due to initial conditions}} \tag{8}$$

Variance Decomposition

Y	percentage of variance due to initial conditions
Lifetime income	7.47%
Lifetime wealth	11.59%
Lifetime utility	4.63%

Table 6: Variance decomposition

Social Mobility

The literature on social mobility estimates process as

$$\log(Y_1) = \alpha + p\beta \log(Y_0) + \epsilon$$

where Y_k is a measure of present discounted value of earnings. In the model, $\beta = 0.43$, in line with empirical estimates.

Quantitative Exercises

- Effect of borrowing constraints
- Effect of intergenerational correlation of ability.
- The rise in college tuition.
- Skill biased technical change.

The Effect of borrowing constraint

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, θ_l)	-7.17%	-14.81%	-14.81%
(a_m, θ_h)	10.3%	20.48%	20.47%
(a_l, θ_m)	-3.78%	-0.24%	-2.5%
(a_h, θ_m)	27.1%	8.8%	25.82%

Table 7: Lifetime Inequality

The Effect of borrowing constraint

Y	% of variance due to initial conditions (baseline)	% variance due to initial conditions g^{ϕ}
Lifetime income	7.47 %	7.49 %
Lifetime wealth	11.59%	12.28%
Lifetime utility	4.63%	5.12%

Table 8: Variance decomposition

The Effect of parental background

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, θ_l)	-6.9%	-15.15%	-13.6%
(a_m, θ_h)	9.5%	20.61%	18.5%
(a_l, θ_m)	-20%	-9.1%	-18.4%
(a_h, θ_m)	39%	30.5%	42.2%

Table 9: Lifetime Inequality

The Effect of parental background

Y	% of variance due to initial conditions (baseline)	% variance due to initial conditions g^{old}	% variance du initial conditi g^{new}
income	7.47%	7.48%	6.89%
wealth	11.59%	11.59%	10.59%
utility	4.63%	4.61%	4.45%

Table 10: Variance decomposition

The rise of college tuitions

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, θ_l)	-7.9%	-15%	-14.66%
(a_m, θ_h)	11.02%	20.50%	20.1%
(a_l, θ_m)	-0.05%	-2.2%	-4.17%
(a_h, θ_m)	53%	30.4%	54.5%

Table 11: Lifetime Inequality

The rise of college tuition

Y	% of variance due to initial conditions (baseline)	% variance due to initial conditions g^{old}	% variance du initial conditi g^{new}
income	7.47%	7.46%	6.67%
wealth	11.59%	11.57%	10.12%
utility	4.63%	4.38%	3.67%

Table 12: Variance decomposition

Skill-biased technical change

(a, h, θ)	Lifetime utility	Lifetime earnings	Lifetime wealth
(a_m, θ_l)	-7.37%	-14.55%	-14.48%
(a_m, θ_h)	10.47%	20.11%	20.02%
(a_l, θ_m)	-1.2%	-0.94%	-1.38%
(a_h, θ_m)	8.09%	1.48%	6.13%

Table 13: Lifetime Inequality

Skill-biased technical change

Y	% of variance due to initial conditions (baseline)	% variance due to initial conditions g^{old}	% variance du initial conditi g^{new}
income	7.47%	7.48%	7.48%
wealth	11.59%	11.59%	11.59%
utility	4.63%	4.63%	4.63%

Table 14: Variance decomposition

Conclusions

Conclusions

- Quantitative macroeconomic model to explain the evolution of cohort inequality over the life cycle.
- It takes into account the interplay of borrowing constraints and parental transfer in the human capital accumulation process.
- Initial conditions shape the observed inequality. About 10% of the variance of lifetime income and wealth is explained by initial conditions.